

Carolina Ignacia Pino Rojas

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RESEARCH INTERESTS

My research interests focus on astronomy and planetary science. I am particularly drawn to the formation and evolution of planetary systems, protoplanetary disks, and exoplanets. I also have a strong interest in stellar astrophysics and astrobiology, and seek to apply scientific computing, data analysis, and machine learning to address astrophysical problems.

EDUCATION

Pontificia Universidad Católica de Valparaíso

B.Sc. in Physics

2025 – Present

Valparaíso, Chile

Pontificia Universidad Católica de Valparaíso

Associate Degree in Sciences (Bachillerato en Ciencias)

Completed: 2026

Valparaíso, Chile

ACADEMIC MOBILITY

Universidad de Atacama

National Academic Mobility Program “Súbete”

2023

Copiapó, Chile

- Completed coursework in Planetary Geology through a national virtual academic mobility program.

RESEARCH EXPERIENCE

University of Chile

Department of Physics

Research Assistant – Summer Research Program

January 2026

Santiago, Chile

- Conducted a systematic literature review by identifying, synthesizing, and comparing international evidence on academic mobility, parenthood, and scientific careers.
- Developed a Directed Acyclic Graph (DAG) describing institutional mechanisms affecting academic career progression through causal modeling.
- Contributed to the development of a modular survey instrument by adapting existing scales, drafting survey items, and proposing indicators tailored to the Chilean context.
- Designed and presented a scientific poster communicating the project’s methodology and findings.

RESEARCH PROJECTS

SIPI AI: Intelligent Flood Prediction System

2026

Team Scientific Computing & Machine Learning Project

Samsung Innovation Campus

- Collaborated in the end-to-end development of a nationwide flood forecasting system integrating hydrological (CAMELS-CL) and meteorological (ERA5-LAND/Open-Meteo) datasets to predict river streamflow and flood risk across Chile.
- Contributed to data acquisition, preprocessing, feature engineering, model development, temporal validation, and performance evaluation using XGBoost for streamflow prediction and extreme flood classification.
- Participated in the development of an interactive Streamlit application for real-time visualization of hydrological forecasts, model predictions, and flood-risk indicators.
- Worked collaboratively using Git and GitHub to develop a reproducible scientific computing workflow, from data preparation and model training to deployment of the final application.
- **Project:** [Website](#)

LEADERSHIP

Pontificia Universidad Católica de Valparaíso

2026

Project Director — “Science or Nonsense?”

University-funded Student Innovation Program

- Led a university-funded science outreach project selected through the *Con tus Ideas Universidad Viva* initiative to promote scientific literacy and critical thinking through gamification.
- Directed project execution by coordinating the student team, assigning responsibilities, managing the project budget, supervising procurement, monitoring timelines, and serving as the primary liaison with the university.
- Coordinated the implementation of pilot activities, educational resources, and public outreach events, ensuring the successful execution of the project.
- Oversaw project organization and logistics for an interactive science outreach initiative targeting secondary and university students.

Fundación Tremendas

2020 – 2023

STEM Volunteer & Co-Coordinator, Academia Atómicas – Astronomical Sciences

Chile

- Co-coordinated the Astronomical Sciences area of Academia Atómicas, contributing to the planning and development of STEM educational activities for girls and young women.
- Mentored students interested in astronomy and STEM, supporting their scientific learning and participation in educational initiatives.

- Collaborated on outreach projects promoting gender equity, leadership, and participation in science aligned with the United Nations Sustainable Development Goals (SDGs).

SCIENCE COMMUNICATION & OUTREACH

Nuevo Foco	2020 – Present
<i>Science Communicator</i>	<i>Chile</i>
– Contributed to science communication initiatives promoting public engagement with science through educational activities and outreach events. – Authored science communication articles for the organization's website and social media platforms, translating scientific topics into accessible content for general audiences. – Conducted interviews with researchers and science communicators, producing outreach content highlighting current scientific research and its societal impact.	
Explora Coquimbo	2021
<i>Invited Panelist</i>	<i>Chile</i>
– Participated in the panel "Academia Atómicas: Reducing Gender Gaps in Science", discussing strategies to promote women's participation in STEM.	
Congreso Futuro	2018–2020
<i>Volunteer</i>	<i>Chile</i>
– Provided logistical support, coordinated public engagement activities, and assisted international speakers during Chile's largest science outreach event.	

HONORS & AWARDS

Silver Honour (Top 10% Worldwide)	2025
<i>International Astronomy & Astrophysics Competition (IAAC)</i>	
Second National Award (Chile)	2025
<i>International Astronomy & Astrophysics Competition (IAAC)</i>	
Global Award Nominee	2020
<i>NASA International Space Apps Challenge</i>	

PROFESSIONAL DEVELOPMENT

Google Data Analytics Professional Certificate	2026 – Present
<i>Google & Talento Digital para Chile</i>	<i>Professional Certificate</i>
– Selected through the competitive <i>Crece con Google</i> scholarship program administered by Talento Digital para Chile. – Currently pursuing the Google Data Analytics Professional Certificate, strengthening analytical and computational skills for data-driven research and scientific problem-solving.	
Samsung Innovation Campus	2026 – Present
<i>Samsung Electronics</i>	<i>Coding & Programming Certificate; Big Data Program</i>
– Selected through a competitive admission process to participate in Samsung Innovation Campus. – Completed the Coding & Programming Certificate and currently pursuing advanced training in Big Data through project-based learning.	
Second Peruvian School of Quantum Computing	2025
<i>QuantumQuipu</i>	<i>Universidad Nacional Mayor de San Marcos, Perú</i>
– Completed advanced training in quantum computing, strengthening theoretical and practical foundations in quantum algorithms, quantum hardware, and quantum machine learning.	
Quantum Winter School	2023
<i>The Coding School & Qubit by Qubit</i>	<i>International Winter School</i>
– Completed introductory training in quantum computing through lectures and hands-on programming activities.	
PROVOCA Mentorship Program	2022 – 2023
<i>AUI/NRAO Chile</i>	<i>National Mentorship Program</i>
– Participated in a national mentorship program focused on leadership, scientific communication, and professional development in STEM.	

COMPUTATIONAL SKILLS

Programming: Python
Scientific Computing: NumPy, Pandas, SciPy, Microsoft Excel
Scientific Visualization: Matplotlib
Scientific Applications: Streamlit, Gradio
Development Tools: Git, GitHub, LaTeX
Languages: Spanish (Native), English (C2 Proficient)